

National University of Computer and Emerging Sciences, CFD Campus



Course Name: Data Science
Program: Computer Science
Submission Date: 26-09-2025
Section: BS(CS)
Exam Type: Assignment-1

Course Code:
Semester: Fall 2025
Total Marks: 100

Instructions/Note:

1. Make single .ipynb file for all Questions and use this format to name it 22F-1234_Task1.pynb.
2. Please submit your file in this format 22F-1234_L1.
3. Do not copy code from any source otherwise you will be penalized with negative marks.
4. Datasets (Employee.csv & employee_training_dataset.csv) is provided with this Assignment.
5. There Will be Viva for this Assignment.

Question No. 1**Marks: 20**

From Employee.csv:

1. Load the dataset and print:
 - Shape, column names, data types
 - Count of missing values per column
 - Count of duplicates
2. Select only the Name, Age, and Salary columns from the dataset.
3. Extract the first 10 rows and the last 5 rows of the dataset.
4. Slice rows where Age is between 30 and 40 and Department is "Engineering".
5. Select all employees from "New York" with Performance_Score above 8.
6. Select the first 5 rows and the last 5 rows of numerical columns only.
7. Identify and fix:
 - Mixed categorical values (e.g., "Male", "M" → "Male").
 - Impossible values (e.g., negative ages, unrealistic salaries).
 - Standardize categorical values.
 - Consistent spelling or capitalization
 - Standardizing abbreviations or codes

Question No. 2**Marks: 15**

1. Handle missing values:
 - Numerical: Fill with mean, median, and compare results.
 - Categorical: Fill with mode or add a new category "Unknown".
2. Detect outliers in Salary and Age using:
 - Interquartile Range (IQR) method
 - Z-score method
3. Treat outliers by either removing or capping them.
 - Show before vs. after summary statistics (mean, median, std).

Question No. 3**Marks: 15**

1. For numerical columns (Age, Salary, Experience_Years, Performance_Score):
 - Calculate mean, median, mode, variance, standard deviation, IQR, min, max, range.
2. Plot distributions:
 - Histograms
 - Boxplots
 - KDE (Kernel Density Estimation) plots
3. Identify which features are skewed and apply transformations (Log / Box-Cox / Yeo-Johnson).

Question No. 4**Marks: 15**

1. Using GroupBy:
 - Find average salary per department.
 - Find average performance score per gender.
 - Find min, max, and mean experience per location.
2. Using Aggregation:
 - Create a summary table with multiple stats (mean, median, std) for Salary and Age by Department.
3. Using Pivot Tables:
 - Create a pivot table showing average Salary by Department and Gender.
 - Create another pivot table showing average Performance Score by Department and Work Location.

Question No. 5**Marks: 15**

1. Encode categorical data:
 - Apply Label Encoding on Gender.
 - Apply One-Hot Encoding on Department.
 - Compare dataset shape before vs. after encoding.
2. Normalize numerical columns (Age, Salary, Experience_Years):
 - Min-Max Scaling
 - Standardization (Z-score)
 - Robust Scaling
3. Show a small table comparing original vs scaled values.

Question No. 6**Marks: 20**

You are given another dataset containing employee training details (**employee_training_dataset.csv**):

1. Join this dataset with the main employee dataset (synthetic_preprocessing_dataset.csv) on Employee_ID.
2. Create a new column showing average salary per training hour.
3. Compare the salaries of employees with Completed vs Pending certifications.